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Non-invasive identification of KRAS mutation in rectal cancer using hybrid intravoxel incoherent motion and diffusion kurtosis model

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Abstract

Background and purpose Sarcoma Viral Oncogene Homologue (KRAS) mutation status is a crucial biomarker in rectal cancer as KRAS mutations confer resistance to anti-EGFR targeted therapies and influence chemotherapy selection, affecting treatment decisions and prognosis. This study evaluates the efficacy of the hybrid intravoxel incoherent motion and diffusion kurtosis imaging (IVIM-DKI) model for non-invasive KRAS mutation identification in rectal cancer.

Methods This prospective study included 73 patients with rectal cancer who underwent MRI scans using an IVIM-DKI sequence. KRAS mutation status was determined through histopathological analysis. The parameters derived from the hybrid IVIM-DKI model, including the apparent diffusion coefficient (ADC), true diffusion coefficient (D), diffusion kurtosis (K), perfusion fraction (f), and pseudo-diffusion coefficient (D*), were compared between the KRAS mutation group and wild-type group. The diagnostic performance was evaluated using the receiver operating characteristic (ROC) curve. The hybrid IVIM-DKI parameters and their association with clinicopathological features were also explored.

Results Significant differences were observed between the KRAS mutation and wild-type groups for ADC, D, and K values ($p < 0.05$). The K value derived from the IVIM-DKI model demonstrated the highest area under the ROC curve (AUC = 0.779) in characterizing KRAS mutation status, with a sensitivity of 88.1% and specificity of 70.3%. The ADC

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value also showed moderate diagnostic performance ($AUC = 0.702$). Specific IVIM-DKI parameters, such as f and K , were associated with various clinicopathological features, suggesting their potential as imaging biomarkers.

Conclusion The hybrid IVIM-DKI model, particularly the diffusion kurtosis parameter, shows promise as a non-invasive imaging biomarker for KRAS mutation identification in rectal cancer. This approach may facilitate early detection of anti-EGFR therapy resistance, potentially guiding personalized treatment strategies and contributing to enhanced clinical outcomes.

Highlights

- The hybrid IVIM-DKI model demonstrates potential as a non-invasive method for identifying KRAS mutation status in rectal cancer patients.
- Diffusion kurtosis (K) value shows the highest diagnostic performance in characterizing KRAS mutation status, with high sensitivity and specificity.
- Specific IVIM-DKI parameters are associated with clinicopathological features, suggesting their potential as imaging biomarkers for tumor characterization and prognosis in rectal cancer.

Keywords KRAS mutation, Hybrid IVIM-DKI model, Rectal cancer

Introduction

Rectal cancer is a common malignant tumor in both males and females and causes a significant public health concern [1]. Kirsten Rat Sarcoma Viral Oncogene Homologue (KRAS) mutation is one of the key factors in its pathogenesis. KRAS mutation can promote the growth of tumor cells, resist chemotherapy drugs, and increase the risk of recurrence [2, 3]. So, the detection of KRAS mutation is of great significance. However, the current gold standard techniques for KRAS mutation detection, such as PCR-based methods and Sanger sequencing, often require invasive procedures to obtain tumor tissue [4].

Functional magnetic resonance imaging technology has been used to evaluate tumor characteristics quantitatively. It can noninvasively evaluate the tissue micro-environment and thus help in cancer characterization. The apparent diffusion coefficient (ADC) value extracted from diffusion weighted imaging (DWI) measures the restricted Brownian motion of water molecules. ADC has demonstrated significant potential in predicting both the pathological staging and the immunohistochemical profile of rectal carcinoma [5]. Recent advances in non-invasive imaging methods for KRAS mutation detection have shown promising results across gastrointestinal cancers. In pancreatic ductal adenocarcinoma, conventional MRI features including rim enhancement and larger tumor size characterize KRAS-mutated tumors, which demonstrate worse prognosis compared to non-mutated cases [6]. In rectal cancer, amide proton transfer-weighted magnetic resonance imaging has emerged as a potential supplement for KRAS mutation evaluation, with mutant tumors exhibiting significantly higher APTw values than wild-type tumors, achieving high sensitivity but limited specificity [7]. Additionally, fractional-order calculus diffusion models have demonstrated superior performance compared to traditional ADC values in distinguishing KRAS mutation status, with combined FROC-specific

parameters showing excellent diagnostic accuracy for KRAS mutation evaluation [8].

Recent studies indicate that, in clinical evaluations of rectal cancer tumors, the bi-exponential model intravoxel incoherent motion (IVIM) and non-Gaussian model diffusion kurtosis imaging (DKI) have higher sensitivity and specificity than the mono-exponential model DWI [9, 10]. IVIM distinguishes between water molecular diffusion and microcirculation or perfusion by separating random molecular diffusion from the pseudo-random motion of blood flow in the capillaries [11]. Although the ADC model assumes that water diffusion in tissues follows a Gaussian distribution, DKI recognizes that this diffusion is often non-Gaussian in biological tissues due to structural heterogeneities such as cell membranes, organelles, and complex tissue architecture [12]. In 2012, Lu et al. extended the IVIM model to account for restricted diffusion by incorporating a measure of non-Gaussian behavior (diffusion kurtosis), enabling the simultaneous estimation of perfusion and restricted diffusion parameters using both IVIM and DKI in head and neck cancer [13]. This method, described as a hybrid IVIM and DKI model, can more accurately characterize diffusion in perfused biological tissues [14]. Because the IVIM and DKI models address different facets of diffusion, a hybrid IVIM-DKI model could provide a more precise characterization of diffusion in perfused biological tissues. Compared with the application of DWI, IVIM, or DKI models alone, this method enhances the model's ability to capture complex details of tumor microstructure [15]. KRAS mutations in tumors affect tumor proliferation and survival and are closely related to the characteristics of the tumor microenvironment [16]. As advanced diffusion models, the hybrid IVIM and DKI models have great potential as non-invasive alternatives for KRAS mutation detection. It can comprehensively characterize the tumor microenvironment by

quantifying tumor perfusion, diffusion, and microstructural complexity.

In this study, we aimed to apply a hybrid IVIM-DKI model to non-invasively identify KRAS mutation status in patients with rectal cancer by correlating diffusion parameters with histopathologically confirmed KRAS mutations, and to evaluate the diagnostic performance of this imaging approach as a potential non-invasive biomarker for KRAS mutation detection.

Methods

Patients

This prospective study was approved by the IRB of Shuguang Hospital affiliated with Shanghai University of TCM (Approval numbers: 2019-750-105-01). Written informed consent was obtained from all participants in accordance with the Declaration of Helsinki (2013

revision). The inclusion and exclusion criteria are outlined in Fig. 1. This study included a cohort of 73 patients.

MRI acquisition

MRI images were obtained using a 3.0T MRI scanner (MAGNETOM Skyra, Siemens Healthcare) equipped with an 18-channel body coil. Patients underwent standardized preparation including at least 6-hour fasting prior to MRI examination. The hybrid IVIM-DKI model sequence employed the echo planar imaging (EPI) technique with the following parameters: repetition time (TR) of 5700 ms, echo time (TE) of 76 ms, flip angle (FA) of 90°, field of view (FOV) of 200 mm × 200 mm, slice thickness of 3 mm, bandwidth of 1724 Hz/pixel, and *b* values of 0, 20, 40, 60, 80, 150, 200, 400, 600, 800, 1000, 1500, 2000, and 2500 mm²/s. Diffusion was acquired in three directions, with a total scan time of 572 s. Routine MRI sequences included axial high-resolution T2-weighted

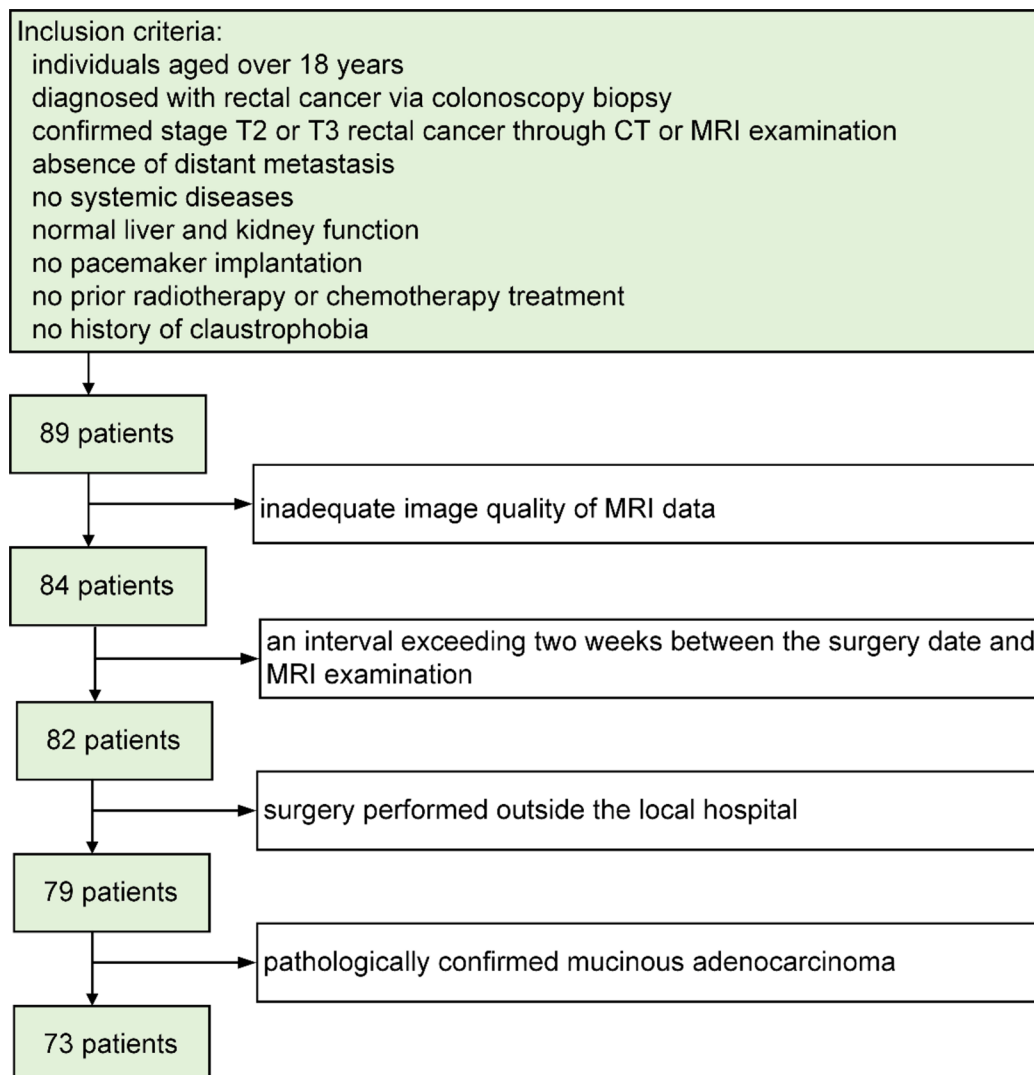


Fig. 1 Flowchart of patient recruitment

imaging (TR/TE of 5300 ms/93 ms, FA of 150°, FOV of 250 mm × 250 mm, slice thickness of 2 mm, bandwidth of 248 Hz/pixel) with a scan time of 138 s. Additionally, coronal T2-weighted imaging, sagittal T2-weighted imaging, and axial T1-weighted imaging were performed as part of the protocol.

Histopathology

Two specialized gastrointestinal pathologists jointly evaluated the pathological features using surgically resected specimens. The staging of rectal cancer follows the American Joint Committee on Cancer Tumor Node Metastasis Staging System [17]. Other histological features of rectal cancer were evaluated, including tumor differentiation degree (classified as well differentiated, moderate, or poor), lymphovascular invasion (LVI), and nerve infiltration (NI). Pathologists perform routine processing on the excised tissue blocks and select tumor areas for KRAS gene testing. Genomic DNA was extracted from tumor tissue sections fixed in formalin and embedded in paraffin using the QIAamp DNA FFPE tissue kit (Qiagen, Chatsworth, CA). The detection of KRAS using the mutation amplification refractory mutation system method is achieved through polymerase chain reaction (PCR) [18].

Quantitative hybrid IVIM-DKI model analysis

Two experienced radiologists with 8 and 15 years of abdominal MRI expertise conducted consensus readings to minimize inter-observer variability. ROIs were manually delineated using polygonal contours on b_0 DWI images at the plane of deepest rectal cancer infiltration. Accurate tumor delineation was ensured by cross-referencing T1WI and T2WI sequences, with high b -value images used to identify solid tumor components. Standardized selection criteria included: (1) targeting the largest solid tumor cross-section; (2) excluding necrotic, hemorrhagic, or cystic areas; (3) avoiding bowel lumen and mesorectal fat; and (4) maintaining a 2 mm margin from tumor boundaries to minimize partial volume effects. The tumor ROI is then replicated on images corresponding to each b value.

The ADC is obtained by fitting all b values to the following Eq. (1):

$$\frac{S_{(b)}}{S_0} = \exp(-b \times ADC) \quad (1)$$

Where $S_{(b)}$ is the signal strength measured at each specific b value (20, 40, 60, 80, 150, 200, 400, 600, 800, 1000, 1500, 2000, and 2500 s/mm²) and S_0 is the signal strength ($b=0$ s/mm²).

Parameter estimation in the hybrid IVIM-DKI analysis is performed using nonlinear least squares (NLLS) optimization and parallel computation along with a custom

MATLAB toolbox (version 9.1, MathWorks, Natick, MA, USA). The analysis utilized all 13 b values from 0 to 2500 s/mm², combining a reconstruction based on the iterated total variational (TV) penalty function with a hybrid IVIM-DKI model.

The parameters of the function are calculated using the signal strength derived from all 13 b values, including true diffusion coefficient (D), and kurtosis (K), perfusion fraction (f), pseudo-diffusion coefficient (D^*). The calculation process is as follows [11]:

$$\begin{aligned} \frac{S_{(b)}}{S_0} = & f \times \exp(-b \times D^*) + (1 - f) \\ & \times \exp(-b \times D + \frac{1}{6} \times b^2 \times D^2 \times K) \end{aligned} \quad (2)$$

In Eq. (2), $S_{(b)}$ is the signal intensity at a given b value, S_0 denotes the signal intensity at $b=0$, and b is the b factor. The Levenberg-Marquardt algorithm was employed to perform the least squares fit of the signal intensities corresponding to the b values. All parameters were derived from the average signal intensity within the ROI. To enhance the fitting accuracy of the bi-exponential model and prevent overfitting, the following procedure was implemented: Initially, the data for $b > 200$ s/mm² were used to determine the parameters D and K . Subsequently, keeping D and K fixed, curve fitting was conducted for f and D^* across all b values using Eq. (2).

Statistical analysis

Statistical analysis was performed using IBM SPSS Statistics version 26.0. Initially, the data were tested for normality. As the data did not follow a normal distribution, non-parametric tests were subsequently applied. The Mann-Whitney U test compared continuous variables, including IVIM-DKI model parameters and ADC values, between the KRAS mutation and wild-type groups. Receiver operating characteristic (ROC) curve analysis evaluated the discriminatory power of parameters showing significant differences between these groups. The ROC analysis provided the area under the curve (AUC), sensitivity, specificity, and optimal cut-off values for these significant parameters. Additionally, the Mann-Whitney U test and Kruskal-Wallis test examined the relationship between IVIM-DKI model parameters, ADC values, and clinicopathological features in rectal cancer patients.

Results

Patient characteristics

Finally, this study included a total of 73 patients. Table 1 provides a summary of the detailed demographic and clinical profiles. The average age of the cohort was 66 ± 11 years. Of the participants, 36 were classified into the KRAS mutation group, while 37 were categorized as the

Table 1 Demographic and clinical features of patients

Variables	Patients
Number (n)	73
Age, years (mean ± SD)	66 ± 11 (37–86)
Sex, male, n (%)	50 (68.5%)
T stages	
T2, n (%)	13 (17.8%)
T3, n (%)	54 (74.0%)
T4, n (%)	6 (8.2%)
N stages	
N0, n (%)	27 (37.0%)
N1, n (%)	23 (31.5%)
N2, n (%)	23 (31.5%)
Tumor grade, n (%)	
High differentiated, n (%)	8 (11.0%)
Medium differentiated, n (%)	44 (60.3%)
Low differentiated, n (%)	21 (28.8%)
Lymph-vascular	
Negative	38 (52.1%)
Positive	35 (47.9%)
Neural invasion	
Negative	40 (54.8%)
Positive	33 (45.2%)
Mutations	
Wild Type, n (%)	37 (50.7%)
KRAS Mutation, n (%)	36 (49.3%)

wild-type group. Table 2 displays the hybrid IVIM-DKI model parameters for both groups, including mean, minimum, maximum, and median, as well as the 25th and 75th percentile values.

Comparison of hybrid IVIM-DKI parameters between KRAS mutant and wild-type groups

Statistically significant differences in ADC, D, and K values were found between the KRAS mutation and wild-type groups ($z = -2.974, p = 0.003$; $z = -2.394, p = 0.017$; $z = -4.105, p < 0.001$; respectively), as detailed in Table 2 and illustrated in Fig. 2. However, no significant differences in ADC, D, and K values were found between the KRAS mutation and wild-type groups the f and D* values

between the groups ($z = -1.137, p = 0.256$; $z = -0.154, p = 0.877$; respectively). Figure 3 illustrates the characteristics of hybrid IVIM-DKI imaging in a patient with wild-type KRAS and a patient with a KRAS mutation. To evaluate the robustness of our IVIM-DKI model fitting, we assessed the curve fitting quality using coefficient of determination (R^2) and root mean square error (RMSE). Representative fitting curves from two patients are shown in Fig. 4.

Diagnostic performance of hybrid IVIM-DKI parameters

The AUC values for ADC, D, K, f, and D* were calculated as 0.702, 0.663, 0.779, 0.577, and 0.511, respectively, as presented in Table 3 and illustrated in Fig. 5. Among these, the K value derived from the IVIM-DKI model demonstrated the highest AUC and accuracy in identifying KRAS mutation status in rectal cancer patients. The ADC value also exhibited satisfactory diagnostic capability. In contrast, the other IVIM-DKI parameters, including D, f, and D*, showed lower AUCs for identifying KRAS mutation status.

IVIM-DKI parameter associations with clinicopathological features

The f value derived from the hybrid IVIM-DKI model showed significant differences across T-stage groups ($h = 6.961, p = 0.031$) and lymph-vascular invasion groups ($z = -2.970, p = 0.003$). The K value also demonstrated significant differences between N-stage groups ($h = 9.917, p = 0.007$) and differentiation groups ($h = 14.314, p = 0.001$). ADC value showed significant differences in the differentiation groups ($h = 7.667, p = 0.022$). In contrast, the D and D* parameters did not show significant differences in any of the clinicopathological features examined. These findings suggest that specific hybrid IVIM-DKI parameters, such as f and K, may serve as potential imaging biomarkers for rectal cancer characterization.

Table 2 ADC values and respective parameters for IVIM-DKI model in KRAS mutation group and wild type groups

IVIM-DKI model-derived parameters	ADC ($\times 10^{-3} \text{ mm}^2/\text{s}$)		D ($\times 10^{-3} \text{ mm}^2/\text{s}$)		K		f (%)		D* ($\times 10^{-2} \text{ mm}^2/\text{s}$)	
	KRAS Mutation	Wild Type	KRAS Mutation	Wild Type	KRAS Mutation	Wild Type	KRAS Mutation	Wild Type	KRAS Mutation	Wild Type
Mean	0.584	0.627	1.234	1.430	0.9351	0.8221	9.160	11.024	0.668	0.728
Minimum	0.518	0.533	0.953	0.672	0.8318	0.5232	2.936	2.361	0.005	0.003
Maximum	0.682	0.825	1.575	2.234	1.0732	1.0775	33.00	29.103	2.872	2.585
25th percentile	0.544	0.582	1.085	1.213	0.8797	0.7247	5.307	5.227	0.010	0.011
50th percentile	0.590	0.621	1.237	1.335	0.9145	0.8306	6.562	10.148	0.220	0.189
75th percentile	0.607	0.653	1.358	1.685	0.9996	0.9002	12.888	16.059	1.516	1.776
z	-2.974		-2.394		-4.105		-1.137		-0.154	
p	0.003		0.017		<0.001		0.256		0.877	

ADC apparent diffusion coefficient, D true diffusion coefficient, K diffusion kurtosis, f perfusion fraction; D*, pseudo-diffusion coefficient

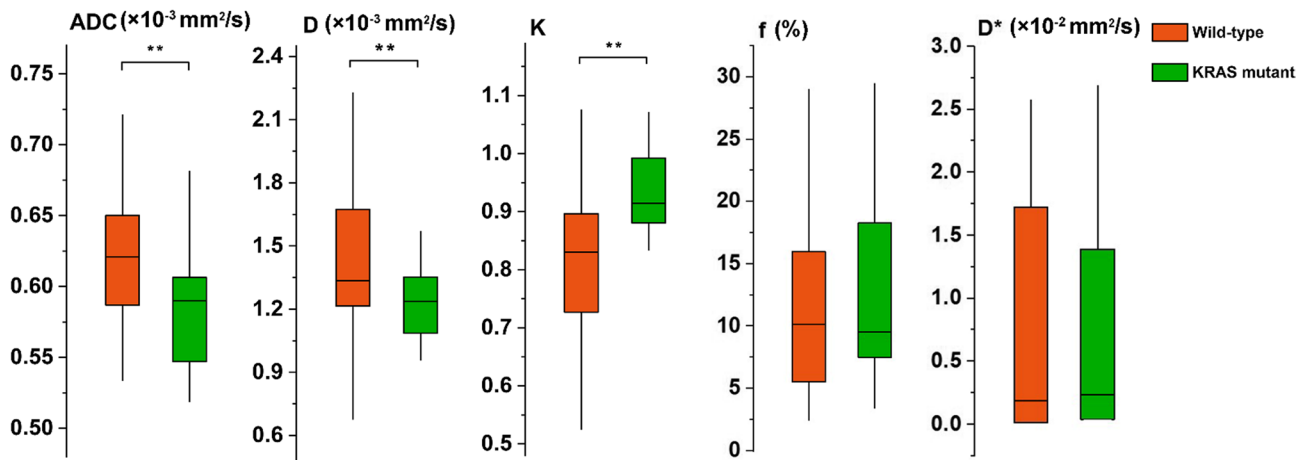


Fig. 2 Boxplots showing the distribution of ADC values and parameters for hybrid IVIM-DKI model in KRAS mutation group and wild type groups

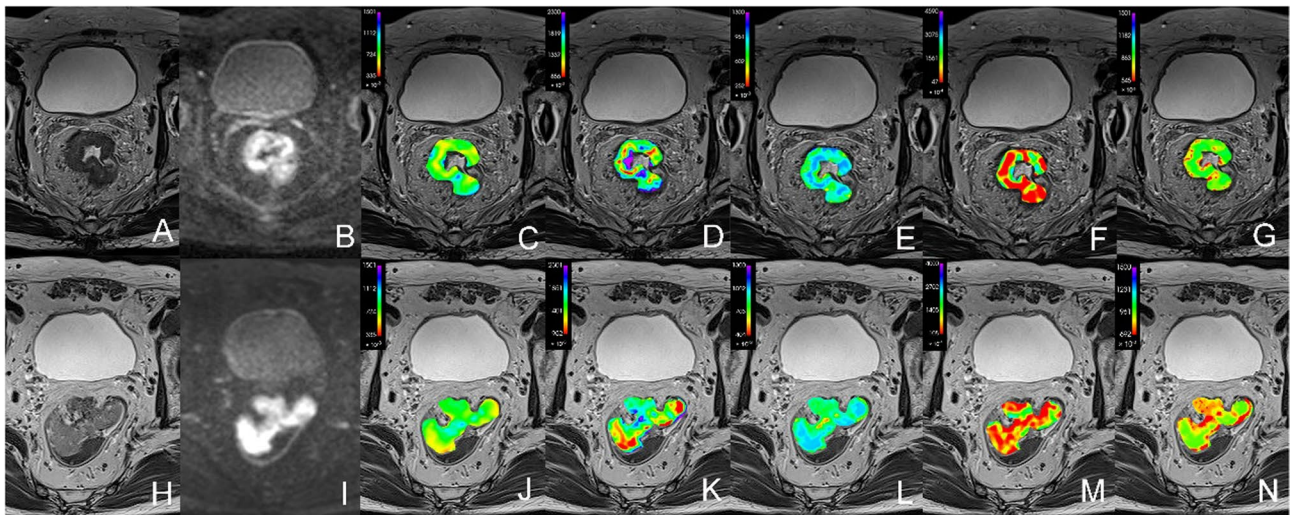


Fig. 3 Imaging of rectal cancer in an 82-year-old male with wild-type KRAS (A-G) and a 58-year-old male with KRAS mutation (H-N). A, H Axial T2-weighted high-resolution sequences planned perpendicularly to the bowel at the tumor site. B, I Diffusion-weighted images obtained at $b = 1000 \text{ s/mm}^2$ (axial). C, J ADC maps. D, K D maps. E, L K maps. F, M f maps. G, N D* maps

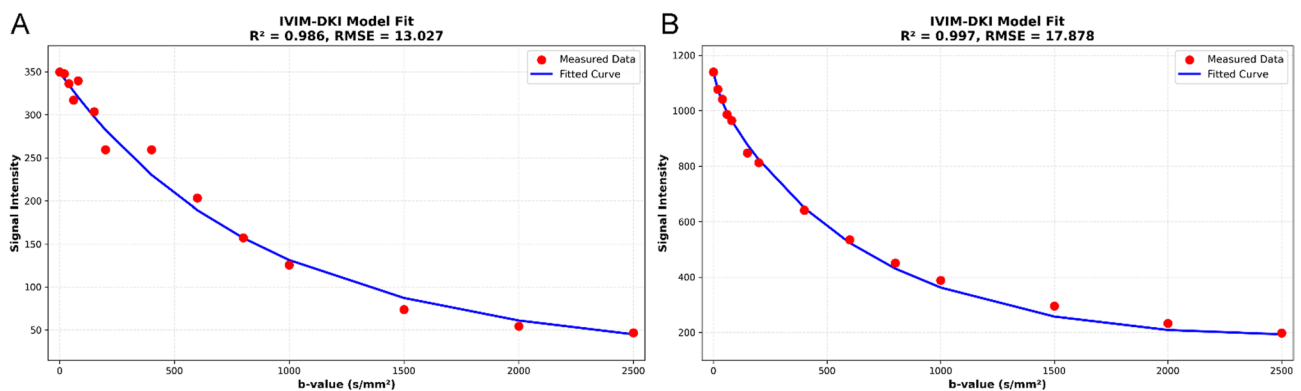


Fig. 4 Representative IVIM-DKI model fitting curves demonstrating fitting quality in rectal cancer patients. A Fitting curve for the 82-year-old male patient with wild-type KRAS (corresponding to patient shown in Fig. 3A-G). Model fitting achieved $R^2 = 0.986$ and $\text{RMSE} = 13.027$. B Fitting curve for the 58-year-old male patient with KRAS mutation (corresponding to patient shown in Fig. 3H-N). Model fitting achieved $R^2 = 0.997$ and $\text{RMSE} = 17.878$

Table 3 ROC analysis results for hybrid IVIM-DKI model in KRAS mutation group and wild type group

	AUC	Cutoff value	Youden index	95% CI	Sensitivity	Specificity
ADC	0.702	0.611	0.373	0.583 to 0.821	80.6%	56.8%
D	0.663	1.476	0.323	0.537 to 0.789	94.4%	37.8%
K	0.779	0.866	0.564	0.668 to 0.891	88.1%	70.3%
f	0.577	9.454	0.318	0.442 to 0.713	75.0%	56.8%
D*	0.511	1.978	0.107	0.376 to 0.645	94.4%	16.2%

AUC area under curve, ADC apparent diffusion coefficient, D true diffusion coefficient, K diffusion kurtosis, f perfusion fraction; D*, pseudo-diffusion coefficient

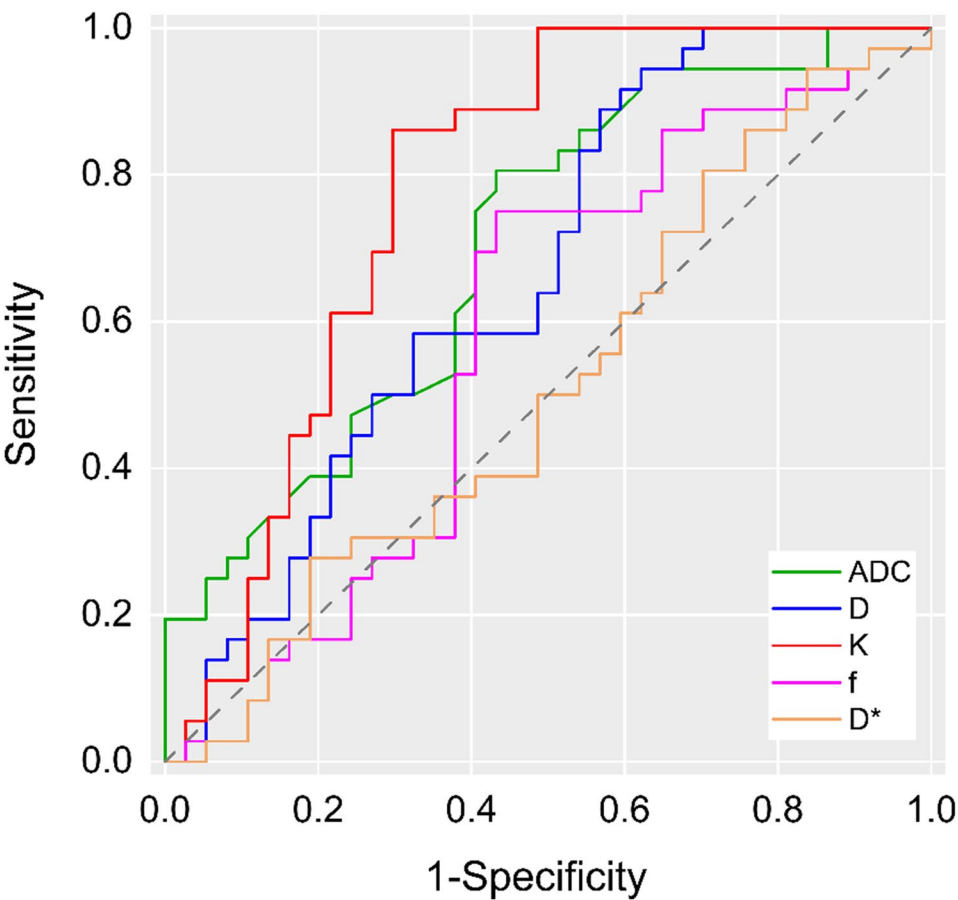


Fig. 5 Receiver-operating characteristic curves for ADC values and corresponding hybrid IVIM-DKI model parameters distinguishing KRAS mutant rectal cancers from wild-type ones

Discussion

The results of this study indicate that the hybrid IVIM-DKI model parameters have important value as imaging biomarkers for identifying KRAS mutations in rectal cancer features. The K value extracted by the hybrid IVIM-DKI model shows the highest diagnostic ability in identifying KRAS mutation status. This indicates that the K value can serve as a valuable imaging biomarker for identifying KRAS mutant rectal cancer, which may help guide clinical decisions by stratifying patients who are less likely to benefit from anti-EGFR targeted therapies.

KRAS mutations are associated with increased cell proliferation, angiogenesis, and metastasis in various cancers, including colorectal cancer [19, 20]. KRAS

mutations are known to affect cellular proliferation, apoptosis resistance, and tumor microenvironment remodeling, which subsequently influence tissue microstructural complexity [21]. The kurtosis parameter (K) quantifies deviation from Gaussian diffusion, reflecting this microstructural heterogeneity at the cellular level. Thus, the association between K values and KRAS mutation status is biologically plausible. Changes in tumor tissue microstructure lead to changes in diffusion properties, making the hybrid IVIM-DKI model, especially the K value, a sensitive biomarker to distinguish KRAS mutants from wild-type rectal cancers. Previous studies of tumor volume histograms have shown that DKI

indicators, especially K_{75th} , are associated with KRAS mutations in colorectal adenocarcinoma [22].

D values and ADC values also showed satisfactory diagnostic power in identifying KRAS mutation status in patients with rectal cancer, in addition to K values. The ADC and D values primarily reflect diffusion parameters, not perfusion [11]. Since the diffusion coefficient is primarily affected by the ratio of tissue to extracellular space, D and ADC are generally regarded as being negatively correlated with cell viability and positively correlated with tissue cystic and necrosis changes [23, 24]. They indicate the extent of water diffusion in tissues and are widely studied as biomarkers for tumor characterization and treatment response assessment [25, 26]. Our findings support the utility of the ADC value in identifying KRAS mutant rectal cancer, likely due to the increased cellularity and reduced extracellular space associated with KRAS mutations. In a DWI study using 8 b values, the maximum ADC, average ADC, and D values in the KRAS mutant group of rectal cancer were significantly lower than the wild-type group [27]. Compared with ADC, the quantitative parameters derived from the hybrid IVIM-DKI model, especially the K value, have a high diagnostic ability in identifying KRAS gene mutations. This indicates that this advanced DWI method has the ability to provide more comprehensive and detailed tumor phenotype information.

Although the K value demonstrated superior diagnostic performance, perfusion parameters (D^* and f) showed no significant differences between KRAS-mutated and wild-type groups, suggesting that diffusion characteristics are more closely related to KRAS mutation status than perfusion properties. The non-significance of f and D^* parameters has important clinical implications. The absence of significant differences indicates that KRAS mutations may not substantially alter perfusion characteristics despite their known effects on angiogenesis pathways. Clinically, this finding suggests that perfusion-based biomarkers alone are insufficient for KRAS mutation detection, emphasizing the importance of microstructural parameters like kurtosis. This has practical implications for imaging protocol optimization, indicating that high b-value sequences capturing microstructural complexity may be more valuable than perfusion-sensitive sequences for KRAS assessment. Non-significant perfusion parameters may still provide complementary information in multiparametric models, potentially improving diagnostic accuracy through machine learning approaches [28]. The biological explanation relates to the heterogeneous nature of tumor perfusion, which is influenced by multiple factors including tumor size, location, inflammatory response, and vascular anatomy beyond KRAS status. KRAS mutations may primarily affect cellular architecture and metabolic processes rather than immediate

vascular changes, making microstructural parameters more sensitive biomarkers. This reflects complex interactions between perfusion and diffusion processes in the tumor microenvironment that vary by molecular subtype.

In this study, we also investigated the relationship between quantitative parameters extracted from the hybrid IVIM-DKI model and clinicopathological features in cancer patients. It was found that the perfusion-related parameter f was significantly different between the T stage groups. This indicates that f reflects information related to blood vessels and angiogenesis in rectal cancer, which is related to tumor invasion and progression [29]. Previous studies have also supported that f reflects microvascular volume fraction, and in studies of esophageal squamous cell cancer, elevated f is associated with more advanced T stage [30].

Our study also found significant differences in K values between groups with N stage of rectal cancer. The K value increased with the increase of N stage of rectal cancer, which may be due to the increased tissue complexity and heterogeneity associated with lymph node metastasis. Earlier research has also highlighted the role of the K value in distinguishing between metastatic and non-metastatic lymph nodes in endometrial cancer [31]. A study employing a hybrid IVIM-DKI model for head and neck squamous cell carcinomas demonstrated that K values could predict the development of distant metastasis [32].

This study found significant differences in ADC and K values between tumor differentiation groups. Poorly differentiated tumors have lower ADC values and higher K values. This may be due to increased cell density in poorly differentiated tumors, limited diffusion of water molecules, and greater tissue heterogeneity. These findings are consistent with previous studies on the use of ADC and K values in evaluating colorectal cancer tumor differentiation [33].

This study also found that f values derived from the hybrid IVIM-DKI model differed significantly between the LVI groups. In tumors with LVI, a reduction in f may contribute to cancer invasion and metastasis. However, this contradicts previous findings in gastric cancer [34]. This may be due to biological and pathological differences between cancers, as well as changes in imaging protocols, patient populations, and analysis techniques.

Several important limitations should be acknowledged in this study. First, this was a single-center study with 73 patients, which may limit the generalizability of findings. However, our sample size calculation demonstrated adequate statistical power to detect clinically meaningful differences in diffusion parameters between KRAS mutation groups. Larger, multicenter studies are needed in the future to validate these findings and assess the generalizability of the results. Secondly, standardization of

hybrid IVIM-DKI acquisition protocols, including optimal b-value selection and fitting algorithms, remains an evolving field. While our imaging parameters were based on established literature recommendations and showed consistent performance, future consensus guidelines would facilitate broader clinical implementation. Third, this proof-of-concept study successfully established the feasibility of hybrid IVIM-DKI for KRAS mutation detection in rectal cancer. While test-retest repeatability analysis was beyond the scope of this initial investigation, future studies should incorporate comprehensive repeatability assessments, including intra- and inter-observer variability analyses, to further validate the clinical robustness of these advanced diffusion metrics for routine implementation.

In conclusion, the results of this study demonstrate the potential of the hybrid IVIM-DKI model in identifying KRAS mutation status in patients with rectal cancer. The ability to noninvasively characterize underlying tumor biology using advanced diffusion-weighted imaging techniques has important implications for personalized management and targeted treatment strategies.

Acknowledgements

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Authors' contributions

J.Y. and S.Z. conceived the study concept and designed the research protocol. J.Y., M.Y., Z.W., W.T., M.L., K.L., and Y.Z. performed the data collection, image analysis, and quality assessment. J.Y., H.Y., and Z.W. conducted the data analysis and statistical interpretation. H.G. and S.Z. provided expert clinical guidance on surgical oncology aspects and critically reviewed the manuscript for scientific accuracy. J.Y. wrote the main manuscript text. All authors reviewed, edited, and approved the final version of the manuscript for submission.

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Data availability

The datasets generated and/or analysed during the current study are not publicly available due to patient privacy.

Declarations

Ethics approval and consent to participate

This study was approved by the IRB of Shuguang Hospital affiliated with Shanghai University of TCM (Approval numbers: 2019-750-105-01). Written informed consent was obtained from all participants in accordance with the Declaration of Helsinki (2013 revision).

Consent for publication

All patients provided written informed consent in the study and consent for publication.

Competing interests

The authors declare no competing interests.

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